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# Designing and Evaluating Games for Landslides, Earthquakes, and Fires: Lesson Learned from Schools in Nepal

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**Abstract:** The Himalayan country of Nepal is vulnerable to landslides, earthquakes, and fires. Its inhabitants need to be empowered on how to react in emergencies to prevent fatalities and respond to crises efficiently while promoting longer-term sustainability and resilience. This research project, a collaborative effort involving teachers, students, and researchers, highlights the design and implementation of games for disaster risk reduction tested in remote schools. Three interactive games were developed using an iterative game design process and testing in workshops aiming to ensure the inclusivity and diversity of the project. The games targeted preparedness and response to landslides, earthquakes, and house fires. The outcome has proven that the game-based approach to teaching and learning is crucial in empowering underserved school children often left out in formal and informal disaster management processes. This study has shown that game-based learning of disaster preparedness and response effectively empowers resource-deficient communities and regions of the Global South.

**Keywords:** disaster risk reduction; game design; analog games; schools; Nepal

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## 1. Introduction

Nepal's unique geographical setting is characterized by steep slopes, tectonically evolving Himalayan ranges, heavy monsoon rainfall patterns, and a diverse spectrum of geological and hydro-meteorological hazards, which presents a complex set of challenges. These hazards, ranging from landslides and debris flows to floods and glacial lake outbursts, along with the threat of earthquakes, fires, drought, and epidemics, underscore the multifaceted challenges faced by the country [1–3]. Recent studies indicate a concerning increase in the frequency of natural disasters. During 2011–2023, 248 earthquakes occurred in Nepal, which caused the deaths of 9131 people [4]. In the same period, Nepal experienced 23,767 fires and 3279 incidents of landslides. These caused the deaths of 3393 people, which consequently affected 44,138 families [4]. This high casualty rate is often attributed to uncontrolled development, environmental degradation, climate change, and human interventions [1,2,5]. The majority of disaster-induced deaths and sufferings occurred among infants, school children, seniors, and other socially vulnerable groups.

These disasters highlight the urgent need for improved disaster preparedness and response efforts to safeguard lives and infrastructure in Nepal. Game-based learning (GBL) offers a promising avenue for the preparedness of school students, enhancing their readiness skills and disaster awareness [6–10]. Studies show that GBL can enhance learning experiences, motivation, and disaster preparedness skills [9,11], improve awareness and response capabilities, and foster civic responsibility [7,8]. Among vulnerable groups, GBL can also enhance the understanding of evacuation procedures and disaster risks [6], significantly improve self-awareness [11,12], and amplify the awareness of evacuation procedures and disaster risk reduction measures [6,9]. Additionally, it may enhance students' crisis

management skills and problem-solving abilities [7,8,10] and foster critical thinking and collaborative behavior. As disasters escalate, integrating games into disaster management strategies becomes pertinent [13]. Games may improve knowledge retention, motivation, and behavioral response during disasters. Game-based activities can often be more effective in teaching about disasters than traditional methods, and students can significantly improve their understanding of disaster-preparedness activities [9]. The relevance of analog games among children is relevant in rural contexts where digital technologies, computer devices, and the internet are unavailable. This study proposes three games for landslides, fire, and earthquake preparedness and response among school children in the rural schools of Nepal.

The existing research on incorporating game-based learning in disaster preparedness primarily focuses on specific disasters. However, there needs to be more uniformity in the population groups targeted for these activities and a noticeable variation in the game design process tailored to specific characteristics of players/users. More comprehensive studies need to be conducted encompassing game design and its evaluation among school children in rural settings. Little is known about how students in rural communities can receive and improve their skills through disaster education games, highlighting the need for more focused research in this area. The lack of analog game options tailored to remote regions exacerbates this disparity, as many rural populations have yet to access digital technologies. This paper aims to bridge this gap in research and concentrates on developing games with teachers in remote schools in Nepal. It describes how three games focusing on landslides, fires, and earthquakes were developed through teacher workshops and tested with students. Thirty-six teachers from eighteen schools in Nepal were involved in the game development. The subsequent sections of this paper outline the research methodology used in the study and discuss the main findings, lessons learned, and future research directions.

## 2. Research Focus

Previous studies have shown that game-based approaches build the capacity for disaster risk reduction. Existing publications have highlighted students' knowledge, skills, and coping capacity for disaster preparedness and response, especially for landslides, fires, and earthquakes.

### *2.1. Game-Based Learning for Disaster Preparedness and Response*

In recent years, game-based approaches have gained momentum for their potential to enhance disaster preparedness and response across diverse contexts. Integrating game-based approaches in disaster education has become a pivotal strategy for enhancing public knowledge, skills, and attitudes toward disaster prevention, mitigation, and risk reduction [13]. Several studies [7,14–16] highlight the positive influence of games in bolstering understanding, comprehension, and engagement in disaster education across diverse audiences. They show the capacity of games as educational tools to enhance players' insights into the complexities of disaster situations and facilitate learning about disaster risk reduction.

Game-based learning has significantly enhanced students' learning experiences and achievements [11,12] and has become a valuable tool in overcoming educational challenges. It can stimulate interest, increase attention span, foster intrinsic motivation, and potentially improve comprehension of concepts and progression in learning. It can enhance knowledge retention, motivation, cognitive abilities, and understanding of concepts included in the learning materials [14]. Games can be dynamic instruments for augmenting disaster risk reduction knowledge and skills and can enhance public awareness [14,15]. Tanwattana and Toyoda [10] and Nopembri et al. [16] emphasized the pivotal role of games in fostering community collaboration and physical fitness. Maulana et al. [17] and Nouchi and Sugiura [18] provided evidence of the effectiveness of games in heightening awareness and improving the response capabilities of people during disasters. Moradian and Nazdik [9]

and Kankanamge et al. [19] stressed the growing demand for innovative disaster education methods and the superiority of game-based approaches in promoting knowledge and behavioral changes in disaster risk management.

Various studies have highlighted the positive impact of disaster-preparedness games on students' behaviors and skills [11,20]. Nopembri et al. [16] and Blasko-Drabik et al. [21] emphasized the role of self-efficacy and cooperative games in enhancing students' disaster prevention skills and stress-coping abilities during disasters. Moradian and Nazdik [9] and Thangagiri and Naganathan [22] demonstrated the effectiveness of games compared to traditional lecture methods in improving students' knowledge and behavior. Tsai et al. [12] and MacDonald et al. [23] highlighted the motivational and practical benefits of disaster-preparedness games. explicitly noting the potential for behavioral change.

For instance, incorporating traditional games and children's songs into landslide response learning has increased disaster awareness and enhanced engagement among elementary school students in West Java, Indonesia [24]. Utilizing a Team Games Tournament (TGT) significantly enhanced youth preparedness and knowledge in dealing with floods and landslides, making absorbing material easier and increasing student participation [25]. Implementing a role-playing learning method assisted by a disaster education pocketbook has proven more effective in disaster mitigation learning than traditional lecture methods, with higher student interest in the learning materials and improved cognitive outcomes in Indonesia [26]. In Taiwan, the design of graphic representation in landslide board games has enhanced the visual communication of ecological concepts, making participants more aware of natural relationships through gameplay experiences [27]. These findings suggest that games can be valuable tools in promoting positive behaviors and skills among students, further advocating for the integration of game-based approaches in disaster education initiatives.

Previous studies have explored the use of serious games to bolster fire preparedness. Mystakidis et al. [28] and Oliva et al. [29] developed virtual reality (VR) games for fire safety training, focusing on school teachers and office workers. Jacob et al. [30] and Mondozzi and Harper [31] targeted the public, emphasizing fire safety awareness and using games to educate children. Safung et al. [32] and Topus and Kerne [33] concentrated on fire evacuation training and emergency response education. Topus and Kerne [33] and Poschner [34] explored using VR and wearable computing in fire brigade training, explicitly focusing on the Paris Fire Brigade.

Many studies have explored the effectiveness of games in enhancing earthquake preparedness. For instance, Nilsen et al. [35] found that video games increased self-efficacy, outcome expectation, and intent to act in earthquake preparedness. Similarly, another study [36] discussed the potential of virtual reality games to train building occupants about earthquake evacuations. Likewise, Winarni et al. [37] developed a mobile educational game for elementary school students, while Chou et al. [38] created a digital adventure game for earthquake escape. Alifia et al. [39] implemented a collaborative, multilingual mobile game for earthquake and tsunami disaster-preparedness education. Alifia et al. [39] and Ersani and Mukminan [40] studied using a disaster mitigation Snakes and Ladders game, and Wu [41] studied the application of game theory in earthquake-preparedness decision-making, emphasizing the game potential in enhancing earthquake preparedness.

Game-based learning may also present challenges despite its many benefits. A significant concern is the lack of methodological rigor of research supporting GBL, which may lead to inconsistent findings related to the effectiveness of games in enhancing knowledge acquisition and engagement [42]. There is a need for more transparency and high-quality evidence. Implementing GBL can be hindered by technical limitations, privacy concerns, and the need for teacher training, which may create barriers to practical use in educational settings [43]. The reliance on games can sometimes overshadow the seriousness of educational outcomes, leading to superficial engagement rather than deep learning [44]. Games may not always ensure an accurate understanding of the material, as students often feel uncertain about the validity of the information they engage with in a less structured

environment [45]. Some teachers may hesitate to adopt game-based methods due to perceived barriers, such as a lack of familiarity with game mechanics and their educational applications [46]. While existing studies [28,30] developed serious games targeting specific groups employing higher-end technologies (i.e., VR, console, applications, and computers) for school teachers and office workers, there is potential to expand the reach of these approaches through simple analog games accessible to all students.

### *2.2. Research Objectives*

The existing literature on game-based approaches for disaster preparedness offers valuable insights into the effectiveness of game interventions, yet several research gaps still need to be addressed. Various studies have shown that games can be developed to enhance earthquake, landslide, and fire preparedness [37,38]. These studies need more coverage of rural contexts, especially in mountainous areas like Nepal, where communities face distinct challenges and vulnerabilities. Many rural schools in Nepal need internet access and more support in the availability of electronic devices such as computers, laptops, or tablets. Exploring the feasibility of analog games for rural areas with limited technological infrastructure can help bridge the gap in access to disaster-preparedness resources and ensure the inclusivity of game-based interventions in very specific rural contexts.

Inclusive game-based participation refers to introducing games that can be equally accessible in all geographical areas and for participants with various technical skills and abilities. To fill these gaps in previous research in game design and game-based preparedness and response to landslides, fires, and earthquakes in the Global South, this study has two main research objectives:

- Develop analog games for disaster preparedness and response employing game-design principles.
- Analyze the perceptions of rural students about disaster education games aiming to improve preparedness and response to landslides, earthquakes, and fires in Nepal.

The following sections summarize research results on developing analog disaster education games with schoolteachers in Nepal. It reviews the process of developing these games and testing them with students.

## 3. Research Methods and Materials

This research involves schoolteachers in game design and testing games with students in their schools. The research methodology consists of the following steps: selecting schoolteachers for game design, designing and conducting game development workshops, implementing prototype games, and revising them after feedback from the students. Figure 1 provides an overview of the research methodology.




**Figure 1.** Research design.

### *3.1. Selection of Schools and Teachers*

As a part of the school disaster risk reduction project in Nepal that was supported by Plus Arts NPO and CITYNET Yokohama Office Japan, 36 teachers from 18 schools were selected for a game design training workshop in 2019 through an open call in five

municipalities of Kathmandu Valley including Kathmandu, Lalitpur, Bhaktapur, Chandragiri, and Shankharapur. A two-step approach was used to prioritize schools' and teachers' participation in the game design process. At first, a meeting was organized with each municipality to brief them about the project objectives and steps. Each municipality was formally requested to invite schools within their jurisdiction for a meeting about game design. Its primary objective was to identify interested and capable teachers who wished to participate in the workshop. After the meeting, each municipality was given criteria and asked to nominate two teachers from their schools for a four-day training workshop. The call explained the criteria and expectations for participation. The workshop's primary objective was to design games to strengthen school students' disaster preparedness and response capacity. Schools and municipalities nominated 26 female and 10 male teachers for the workshop. All selected teachers were teaching in grades 4 to 9. Figure 2 shows the location of the 18 selected schools participating in the workshop.

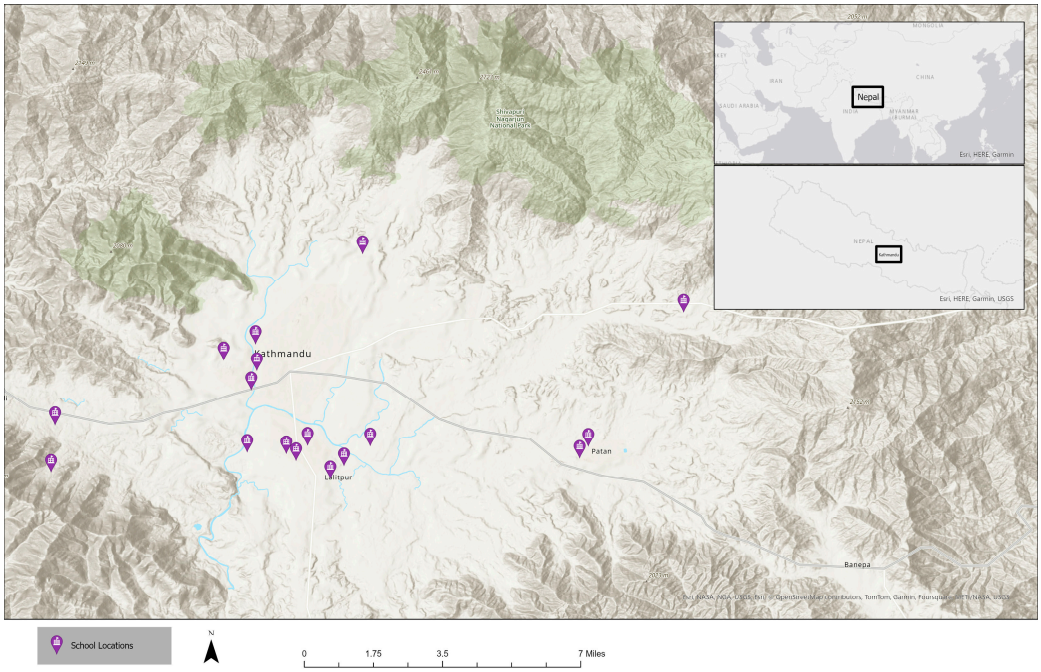



**Figure 2.** Location of selected schools for game design workshop.

### *3.2. Game Development Workshops for Teachers*

The workshop was held in the Lalitpur Municipality of the Kathmandu Valley. Its primary objective was to design games for disaster preparedness and response. The workshop was facilitated by a disaster education expert from Japan, a game design expert from Thailand, and the lead author of this paper.

The workshop's motivation was to develop disaster education games for the students using local knowledge and familiar games within Nepal. The agenda was divided into brainstorming, disaster prioritization, and game design and sharing. The brainstorming took place for the whole group who were given prompts related to disaster prioritization and game selection. Discussions on different disasters took place, and in the end, three disasters were selected through several iterative discussions among the teachers. The three chosen disasters were landslides, fires, and earthquakes. The second part of the brainstorming was related to game design. The teachers identified three popular games in Nepal including Snakes and Ladders, Relay, and Picture Story. The themes of the game design were aligned with the selected disasters: earthquakes, landslides, and fires, as these were the major concerns raised by the teachers during the municipal meetings and the first brainstorming session in the workshop.

Participants were then divided into three groups for the game design part of the workshop. The first group represented Kathmandu Metropolitan City, the second one Lalitpur Metropolitan City, and the third group represented the remaining three municipalities, including Bhaktapur, Shankharapur, and Chandragiri. There were six schools in each group. Among the three disasters, the Kathmandu group selected fires, the Lalitpur group selected earthquakes, and the third group selected landslides. Due to the congested urban settlement, the residents of Kathmandu are at high risk of fire; the teachers of Lalitpur chose earthquake as all the participant schools were located in the historic core area of the City near Darbar Square, which is at high risk of earthquake damage due to the historic buildings built prior to construction codes. From the third group, two municipalities, Shankharapur and Chandragiri, are located in the hills and frequently face landslides.

After finalizing the themes, each group identified local games played by the students and was tasked with integrating the disaster theme into the selected game. The first group of teachers from Kathmandu chose the storytelling method. The Lalitpur teachers decided to work on a Relay Race for Earthquake Awareness. The teachers from the other three municipalities selected the Snakes and Ladders game. The groups' efforts to streamline existing games with themes of the selected disaster continued for two days. During the first half of the first day, they explored the principles of game design. In the second half, they started discussing, brainstorming ideas, and streamlining games (Figure 3). On the second day, they fully engaged in developing their versions of the games while receiving rigorous feedback from the game design expert. In the first half of the third day, they completed the draft of the games and prepared presentations.



**Figure 3.** Teachers of Kathmandu engaging in creating a Picture Story Game for Fire Safety (**left**); Teachers of Lalitpur engaging in developing a Relay Game for Earthquake Awareness (**right**).

On the second half of the third day of the workshop, the three games were presented to the whole group, and feedback was collected from all participants, teachers, and experts (Figure 4). After the feedback from participants and experts, each game underwent further revisions.

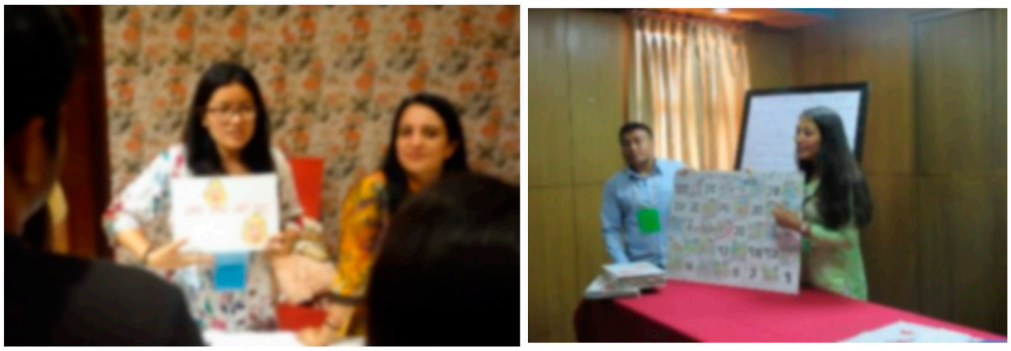


**Figure 4.** Teachers present Picture Story Games for Fire Safety (**left**) and a Snakes and Ladders Game for Landslide Awareness Building (**right**) games in the workshop.

On the fourth day of the workshop, an event was organized at a school in Kathmandu to test these games. At this event, participating teachers demonstrated the three developed games to 70 students in grades 6 and 7 (Figure 5). Seventy students were divided into seven groups (ten in each) and played these three games. Three groups participated in the games at a time, and the remaining four groups acted as observers. Once the first three groups completed playing all three games by taking turns, the other groups participated. After all groups participated in these three games, all the participants were gathered in a playground for a reflection session. During the session, semi-structured questions were asked of the group. Questions were asked to gather the user experience of a particular game to gather feedback on their understanding of each of the three games, the game rules and parts, and disaster preparedness and response. All students were asked to mention their names and grades when they provided the feedback. The semi-structured questions included:

- How did you assess the playing instructions for the Snakes and Ladders game, Relay game, and Picture Story show? Were they clear? Do you have any recommendations to make them more straightforward?
- Which skills have you learned from three games?
- What did you understand about the landslides from the Snakes and Ladders game, about fire preparedness from the Picture Story, and about earthquake awareness from the Relay race?
- What are the difficulties you faced during playing these games?
- Did you struggle with the text and pictures in the games?
- Are the texts and images of the games sufficient for playing?
- Is the message of each game easily understood?
- What changes are required in the games to make them easy to play?
- Do you have any feedback or suggestions on these games?



**Figure 5.** Teachers present Picture Story Games for Fire safety (**left**) and a Relay Game for Earthquake Awareness (**right**) to the students.

After the feedback from the session with the participants, some game elements were adjusted to make them easier for students to understand and follow. Table 1 summarizes the feedback and necessary adjustments in the games. Students mentioned that the images in the Snakes and Ladders Game for Landslides Awareness Building (Figure 6) were unrealistic, and some sixth-grade students mentioned that the text was difficult to understand. Some students noted that the technical jargon was not easily understandable, including disaster preparedness risk and resilience. They were changed by replacing images and language with realistic images and day-to-day images of the landslide scenario. In the Picture Story Game for Fire Safety, the students mentioned that it would help them learn better if there were wrong pictures, prompting them to decide whether it was right or wrong. Another comment was made about providing reasoning for each right and wrong answer for the Picture Story Game for Fire Safety. Both comments were addressed in the next iteration of the game. Finally, students identified the need for multiple answer cards

for each Relay Game for Earthquake Awareness prompt. This issue was also addressed in the next version of the game. Further detailed comments are presented in Table 1.

**Table 1.** Students' feedback on game design and rules (total participants = 70).

| Game | Students' Comments | Changes Made in Each Game |
| --- | --- | --- |
| Snake and Ladder Game for Landslides Awareness Building | • Images that showed the mountain region were not realistic.<br>• Images of dead animals were not realistic.<br>• The texts included in the game were difficult to understand and jargon from disaster management field was not well understood by the students) | • Realistic context-relevant images from Kathmandu Valley were included in the game.<br>• The following changes were made (The number represents the square box number of the Snake and Ladder)—2. before going 5. Housing area/damage, 8. Forest, 9. Livestock, 13. Forest, 16. Livestock, 19. Crisis oriented, 23. Transport safely away from human habitation, 32. Safe, 33. Sleep slope, and 37. Building construction and rapid population. |
| Picture Story Game for Fire Safety | • To make the game more interactive, it is better to provide a picture that indicates an incorrect action in the story and ask students whether this is a correct or incorrect action.<br>• Reasoning is required for students to make each correct or incorrect decision. | • Three pictures of incorrect actions were added to the story; the first picture was about a boy sleeping without turning off the candle; the second picture was about walking in the smoke-filled room; and the third one was touching the door handle directly.<br>• Reasoning for each correct and incorrect picture selection was explained from the lens of fire preparedness and response. |
| Relay Game for Earthquake Awareness | • Multiple answer cards should be offered for each question, allowing each participant to select one at a time.<br>• There were 30 answer cards, which were not sufficient to select option cards in a very short time.<br>• Question no 1 and question no 4 of the A3 size question card were the same; they needed to be changed; the repeat question was: What kind of food should be eaten when staying in a group after an earthquake? | • Five option cards were prepared for each of the 10 questions. Once additional cards were developed, the total number increased to 57 from 30.<br>• The repeated question was changed as follows: How should you eat in a group after the earthquake? |




**Figure 6.** An analog (**left**) and digital (**right**) prototype of the Snakes and Ladders Game for Landslides Awareness Building (The text is written in Nepali language, where various disaster preparedness activities are shown through snake and ladder, and the name of the game is written in the local language, i.e., in Nepali on the top left and written on English in Top right of the game).

### *3.3. Digitization, Distribution, and Evaluation*

After receiving student feedback and fine-tuning the games, the three games were digitized and printed. The digitization process was meant to convert materials that were initially developed using pencils, colors, and other local materials into printed versions with the support of a designer.

In 2022, the first author of this paper introduced the three game prototypes, the Snakes and Ladders Game for Landslides Awareness Building, Picture Story Game for Fire Safety, and Relay Game for Earthquake Awareness, in eight different schools in the Barpak Sulikot rural municipality of Nepal, which is located in the Gorkha district. This

area was the epicenter of the 2015 magnitude 7.8 earthquake and was heavily impacted, including collapse and significant damage to all local schools. The schools were selected based on the recommendations of the education department of the Rural Municipality. Orientation training was organized. One of the local organizations in Gorkha supported the coordination with municipalities and sponsoring the training. Sixteen teachers from eight schools were trained for two days on the importance of disaster risk reduction education in schools. The three games were introduced to teachers during this training. The content was about game-playing steps and procedures, delivering the learning to students, and providing the concept of disaster preparedness through a games-based approach to students in grades six to ten.

After the training, the games were distributed to all eight schools. Two schools were selected to evaluate the games' effectiveness in understanding preparedness for three disasters. These two schools were selected randomly to represent two earthquake-impacted wards of the rural municipality. After receiving the training and game materials, the trained teachers selected twenty students from their schools, representing grades 6 to 10. The 40 selected students played the three games between June and September 2022. At the end of September 2022, focus group discussions with the students from both schools were conducted to assess their understanding of the games and their effectiveness. All 40 students were gathered in a hall, and discussions were recorded on a tape recorder. Equal opportunities were given to each student to answer the questions. Some of the discussion-guiding topics were students' learning from the games about the understanding of landslides, earthquakes, and fires, mitigation strategies at individual and household levels before these disasters, fire safety measures, reasons for considering fire a friend or foe, and escaping from the smoky indoor environment in case of fire. The discussion also captured the clarity of the game rules, expected outcomes, and players' roles. After this feedback, the games were finalized and are presented in the following section.

## 4. Results and Discussions

This section summarizes the final prototypes of the developed games in the workshop, including the Snakes and Ladders Game for Landslide Awareness Building, the Picture Story for Fire Safety, and the Relay Race for Earthquake Awareness. This section explains the game environments, actors, rules, goals, players, and lessons learned from each of the three games developed for grades 4 to 10.

### *4.1. Snakes and Ladders Game for Landslide Awareness Building*

#### 4.1.1. Game Environment

Snakes and Ladders is a popular game among Nepalese children. It is played on a large board featuring squares arranged in a grid. It contains 40 squares with information related to landslide preparedness and response. The board has colorful illustrations and symbols about landslides, making it visually engaging and informative. It measures 28.5 cm by 24.5 cm. Figure 6 shows the analog and digital versions of the game.

#### 4.1.2. Game Actors

The main actors in the game are the players, usually children, who navigate the board using tokens and a die with one to six dots. The game metaphorically represents elements of landslides through snakes and ladders. Educators and facilitators guide the players and explain the educational content.

#### 4.1.3. Rules

Players take turns rolling the die. Depending on the roll, they move their tokens forward on the board. Landing on a ladder allows the player to climb to a higher square representing positive actions like afforestation or proper road management. Landing on a snake causes the player to slide down to a lower square, symbolizing negative outcomes

such as building houses near rivers or ignoring weather updates. The game continues until a player reaches the final square.

#### 4.1.4. Players

The game is designed for 2–4 players. Each player uses a distinct token, such as colored dots, mini wheels, or coins. The players represent individuals learning about landslide preparedness and response. Their roles involve making decisions based on the information provided in each square and understanding the consequences of those decisions.

#### 4.1.5. Goals

The primary goal is to reach the last top square on the board, which signifies a high level of preparedness and understanding of landslides. Players learn about preventive measures and the consequences of neglecting safety protocols. The secondary goal is to foster a collaborative learning environment where players discuss and reflect on the information encountered during the game.

#### 4.1.6. Learning in the Game

The Snakes and Ladders game for Landslides Awareness Building aims to engage students in interactive ways regarding the causes and consequences of landslides. Learning objectives in this game include:

- Awareness Building: Players may better understand how human activities can contribute to landslides and the importance of preventive measures. Different square boxes show how human activities can be responsible for inducing landslides, such as building homes on steep terrain, building roads haphazardly in hill areas, and the consequences of deforestation. These concepts ultimately increase students' awareness about the causes of landslides.
- Practical Knowledge: Students can learn practical strategies for mitigating landslide risks such as afforestation, environmental impact assessment conducted before road construction, and emergency preparedness.
- Enhanced Engagement: The game's cultural significance and familiar format enhance students' engagement and retention of the educational content.
- Collaboration and Reflection: Through gameplay, students can discuss and reflect on the information, which promotes collaborative learning and critical thinking.

### *4.2. Picture Story Game for Fire Safety*

The main goal of the Picture Story Game for Fire Safety is to teach students how to use fire correctly with precautions. It is implemented as a visually engaging narrative with pictures to convey a story and encourage active student participation. It is told through illustrated cards and is driven through a character narrative. It teaches diverse methods to prevent fires at homes by engaging students with a narrative encouraging a heightened awareness of fire safety measures. Figure 7 shows the analog and digital versions of the game.




**Figure 7.** An analog (**left**) and digital (**right**) Picture Story Game for Fire Safety prototype (On the digital picture (right side) the first card is the heading of the story written in Nepali, which means Fire: Friends or Enemy; in subsequent cards numbers from 1 to 10 are written in Nepali language).

#### 4.2.1. Environment

The environment of the Picture Story Game for Fire Safety is represented by a set of A3-sized cards containing ten different pictures (Figure 7). Each card depicts different stages of a story related to fire safety. It can be played indoors or outdoors. An indoor environment, such as a classroom, allows for structured discussion, while an outdoor setting can provide a dynamic and engaging experience for larger groups.

#### 4.2.2. Game Actors

The main character in the game is a schoolboy whose lifestyle resembles most rural Nepalese students. The story is about his daily routine after he comes home from school and completes his homework when a fire incident occurs due to negligence.

#### 4.2.3. Rules

The game involves a sequence of illustrated cards that tell a comprehensive story about fire safety. The educator shows each card to the students, narrates the corresponding part of the story, and asks questions to prompt discussions. The story presents various scenarios, including those related to the possibilities of actual fire incidents and post-fire implications. They are accompanied by incorrect actions. Students must identify these mistakes and suggest correct responses. The mistake cards are shown, such as the boy falling asleep without turning off the candle, standing up and trying to come out after the room filled with smoke, and touching the door handle with bare hands without using any precautions. Key lessons include crawling out to avoid smoke inhalation and following the "Stop, Drop, and Roll" technique [47,48] if their clothes catch fire. The game emphasizes active participation, critical thinking, and practical applications of fire safety knowledge.

#### 4.2.4. Players

The game is designed for 6–10 players but can be adapted for individual players or bigger groups. Each student plays the role of a participant who interprets the images, engages in discussions, and provides their perspectives. The educator acts as the facilitator guiding the narrative and providing feedback.

#### 4.2.5. Goals

The primary goal is to educate students about the correct and safe use of fire while focusing on practical strategies to prevent and respond to fire emergencies. Secondary goals include developing critical thinking and reasoning skills, fostering an awareness of fire prevention methods, and encouraging students to apply their knowledge in real-life situations.

#### 4.2.6. Learning in the Game

The Picture Story Game for Fire Safety aims to achieve the following educational objectives:

- Increased Fire Safety Awareness: Students learn about the importance of using fire safely and the potential dangers of neglecting safety protocols. It is through the story that they can relate to the fire scenario, their daily lives, and their familiarity with prevention measures such as turning off the candles or kerosene lamps when they are not in use, crawling down on the floor when the room is filled with smoke, and not touching metal objects directly with their hand during their escape of fire incidents that may occur in their home.
- Critical Thinking and Reasoning: The game encourages students to think critically about each scenario, identify mistakes, discuss their strategies, and provide reasoned answers.
- Practical Knowledge: Through interactive storytelling, students learn about fire prevention methods such as extinguishing candles and kerosene lamps before going to bed at night, maintaining fire safety measures at home, keeping emergency kits handy,

and using appropriate techniques to protect themselves from smoke and fire during the incident.

- Engagement and Participation: The visually engaging narrative and interactive format enhance student engagement, making learning enjoyable and memorable.
- Application of Knowledge: Students can be better prepared to apply their fire safety knowledge in real-life scenarios by relating to the main character and imagining themselves in similar situations.

The game's character-driven narrative, visual storytelling, and interactive elements effectively teach diverse methods to prevent home fires. This game is a valuable educational tool for raising student awareness about fire safety.

### *4.3. Relay Game for Earthquake Awareness*

The relay game is popular among kids in Nepal. In a relay race, players participate as a team rather than as individuals. The key element of a relay race is the baton. The first runner of each team begins the race with a baton. Once the first runner completes their leg of the race, the baton must be handed off to the team's next runner. This process continues until the final runner, holding the baton, completes the race.

Figure 8 shows the analog and digital implementation of this game. The relay game aims to help children overcome their trauma and prepare them for future earthquakes in their schools and homes in rural areas of the country. It includes various ways of dealing with earthquakes before, during, and after the disaster.




**Figure 8.** An analog (**left**) and digital (**right**) prototype of the Relay Game for Earthquake Awareness (The theme of the game is written in Nepali on the first card, which means earthquake awareness, followed by two question cards with question number and question. The first question shown in the picture is How to eat food in a group after an earthquake? The second question shown in the picture is What do we need to do if we are inside the home during an earthquake?).

#### 4.3.1. Environment

The game's environment is represented by 57 cards featuring images and instructions, representing different obstacles, such as damaged houses and vehicles on the road, and 10 questions related to earthquake preparedness and response.

#### 4.3.2. Game Actors

The primary actors in the game are two students who play the role of a crack house and a car. The students of each team participate in the relay race. The facilitator of the game asks questions and evaluates the answers. The students take on dual roles as runners and obstacles.

#### 4.3.3. Rules

- Team Formation: Participants are divided into two groups of five to eight members forming two lines. Each participant is assigned a number.

- Game Setup: A table is set up with cards featuring images of obstacles such as moving cars and earthquake-damaged houses. Two players from each team act as obstacles based on the instructions on the cards.
- Question and Answer Process: The facilitator poses earthquake-related questions using big question cards. After a bell rings, players must race to the table, navigate the obstacles, find the correct answer card, and return to the last row of their group.
- Turn-taking: Only one member can attempt to find an answer at a time and hold onto their chosen card without exchanging it. Each player has a 30 s time limit to find the answer.
- Scoring: The facilitator assigns one point for each correct answer card and verifies the answers to ensure accuracy. The team with the highest score at the end of the game is declared the winner.

The game starts with the facilitator explaining the rules and the importance of earthquake preparedness. The facilitator poses questions related to earthquake safety. Players navigate obstacles such as moving cars and damaged houses symbolizing real-life challenges during an earthquake. They race to the table to find the correct answer card and return to their team. The game covers all three stages of an earthquake: before, during, and after, with ten questions posed by the facilitator.

#### 4.3.4. Players

The game is designed for groups of five to eight students, each forming a team. Roles in each group include:

- Runners: Team members who race to find the correct answer cards.
- Obstacles: Two players from each team act as moving obstacles to make the game challenging.
- Facilitator: The person who poses questions, oversees the game, and evaluates the answers.

#### 4.3.5. Goals

The primary goal of this game is to win the race by bringing correct answer cards to the questions. This educates students about earthquake preparedness and response strategies before, during, and after an event. Secondary goals include promoting teamwork, quick thinking, and strategic decision-making.

#### 4.3.6. Learning in the Game

The Relay Game for Earthquake Awareness empowers and builds capacity among school students in the following ways:

A Comprehensive Earthquake Knowledge: Players gain a thorough understanding of preventive measures and safety strategies for earthquakes as they must identify the correct answers to the different questions related to earthquake preparedness and response. Teamwork and Collaboration: The relay race format promotes concentration, teamwork, and player cooperation. Quick Thinking and Decision-Making: The time-bound nature of the game encourages quick thinking and strategic decision-making. Engagement and Participation: The dynamic and interactive format ensures that students are actively engaged and absorb the information effectively. Preparedness and Resilience: Students can be better prepared to respond to an earthquake by participating in the game, making them more resilient in real-life scenarios. The game combines education and physical activity, making it an effective and engaging tool for teaching students about earthquake safety. Through teamwork and interactive gameplay, students can learn essential strategies for staying safe during earthquakes and develop a sense of preparedness and resilience.

### *4.4. Lessons Learned from Playing Games with Students*

After designing, implementing, and testing these games, lessons were learned that may help improve the current game prototypes in the next phase of development.

#### 4.4.1. Snakes and Ladders Game for Landslide Awareness Building

Students showed high interest in the Snakes and Ladders Game for Landslide Awareness Building. They found the game exciting and engaging. It has a fun component as the ladders symbolize success and the snakes symbolize failure. Students shared that they always wanted to skip the mouth of the snakes as it made them fall and gave them less chance to win the game. One of the students from grade 7 said: *We often play this game at home, but for the first time, we found out that we can learn about landslides from the game. We understand that landslides occur due to human activities, and we need to be responsible to control this. He further added that we should stop making roads haphazardly and need to plant trees in sloppy areas.* The game was familiar to students. They rated it as fun and informative. While explaining the relevance of the game to enhance their awareness, preparedness, and skills, another student shared: *Landslides are common in our area, and through this game, we learned that if we cut one tree, we need to plant two trees; we also learned that we always need to make a Jhatpat jhola (emergency bag) and prepare for landslides.* Some students reported learning different aspects of landslide mitigation. Others shared that the content was already familiar to them, as they knew about landslides and their mitigation strategies.

Previous research shows that the Snakes and Ladders game has been effectively utilized as an educational tool for disaster mitigation, particularly in enhancing students' knowledge and preparedness for various disasters [40,49–51]. Research indicates this game can significantly improve students' understanding of landslide and earthquake disaster mitigation. Ward [46] showed that students who engaged with the Snakes and Ladders game demonstrated a 67.65% improvement in earthquake preparedness compared to a control group, highlighting its effectiveness in fostering active learning and engagement. Kusumaningjati et al. [49] report students' increased knowledge about landslide disasters, making learning more interactive and enjoyable through the implementation of the game.

#### 4.4.2. Picture Story Game for Fire Safety

Students found this game interesting and relevant because it reflected the scenario already experienced in their community. During the focus group discussion among the students from the two schools who participated in all three games, a sixth-grade student shared: *Through the picture story, I could learn about fire prevention through memory and logical sequences.* The second student from grade seven added: *I understand the concept of staying close to the ground when there is smoke in the room. Smoke normally rises, so we need to stay low to prevent inhaling it.* The third student reported learning as follows: *They should not touch the door handle directly if there is a fire in the room as their hand may get burned; the handle may be made of metal, and the metal is a good conductor of the head.*

Students shared that they know the concept of stop, drop, and roll, which is necessary if they catch on fire. Some of them mentioned that they liked the options and processes presented in the game. The best part of this game was that they learned how to escape from the room full of smoke. They explained that the scenario may be less relevant in the present context as there is a minimal electricity outage, and some alternatives to candles are being used. A more modern scenario could include battery-operated flashlights or the chargeable flashlights that are used nowadays. They also mentioned an increasing number of forest fires and house fires for reasons other than candles or kerosene lamps, and they would be interested in learning about ways to minimize such disasters.

#### 4.4.3. Relay Game for Earthquake Awareness

Students responded positively to the game, enjoying its various components: teamwork, timing, race, and memory. The physical movements and navigating obstacles were considered exciting components of the game. One student shared: *It is delightful and a teamwork game; we need to decide very quickly and select the answer we like the most; though we know how to answer, it is tough to get the exact card we are looking for during a short period. Ultimately, we learned we must avoid cracking houses and cars during an earthquake. We also*

*learned that when we are at home, we must follow the duck, hold, and cover methods. We need to use a pillow to cover our heads, and we can stay under the door's beam during an earthquake.*

Another student said that she enjoyed playing the game, especially because she liked that they needed to engage children in different activities like playing, dancing, and singing after an earthquake. Another student said she liked sharing food due to the limited resources after an earthquake when people stay together. Some students noted that they learned about the planning beforehand, such as making emergency bags, learning first aid procedures, identifying open spaces, etc.

These findings are also supported by previous research on physical games on earthquake preparedness. Physical games and simulations play a crucial role in earthquake preparedness by providing interactive and engaging learning experiences. Studies have shown that earthquake drills are the most effective training method for teaching protective behaviors, especially compared to digital games and traditional education. Earthquake drills allow participants to physically rehearse actions leading to better retention and understanding of the necessary steps during an earthquake [52]. Moreover, games designed for seismic risk education highlight the importance of collaboration over competition. These games successfully engage participants and instill best practices in seismic risk management, particularly in multicultural and multilingual settings [53].

## 5. Conclusions and Future Research Directions

This article addresses an emerging need to educate students and youth in rural Nepal about major disasters, their possible consequences, and actions in terms of preparedness. Game-based learning effectively engages children and youth in the most sensitive and timely issues of modern times [54,55]. The risks of disasters are growing globally for geological and hydro-meteorological events, especially among rapidly transforming economies.

### *5.1. Conclusions*

1. This study has shown that game-based approaches can engage and empower the youth and children to be ready to face the challenges posed by these events. Given the rise of digital and manual games in recent decades, they are always ready to play games. Children and youth are historically the most vulnerable to disasters [56–58]. The three games developed in this study are excellent foundations for building disaster risk reduction capacity among children and youth. They were developed and tested in two schools of Gorkha, an epicenter of the 2015 Nepal earthquake. The experience with all three games showed that games improved students' knowledge and skills in preparedness and response to landslides, fires, and earthquakes.
2. Storytelling and visual elements are powerful disaster risk reduction (DRR) education tools. By making information accessible and engaging, Picture Story games can enhance understanding and preparedness, as evidenced by studies conducted in Japan and other regions. It has been effective in Nepal, where financial, technical, and institutional resources for disaster risk education are limited.
3. Rural Nepal, which grapples with geographic isolation, administrative incapacities, and inadequate infrastructure, is especially vulnerable to multiple disasters. Locally developed, context-specific games empower students to respond effectively to emergencies, bolstering rural resilience.
4. The use of pictures and storytelling is understudied in the context of disaster management. The three games developed during this study have been in the Nepalese context for a long time. It was a novel finding that popular games among children could be used to empower schoolchildren. This research has proven that game-based methods effectively develop knowledge and skills of disaster risks and preparedness among Nepal's schoolchildren, one of the country's most vulnerable groups.
5. Disaster preparedness can be effectively enhanced through picture story games, as evidenced by various studies. For instance, a study in Japan emphasizes the importance of inclusivity and fun in learning about disaster risks, particularly for diverse populations

like foreigners. Tsuda and Yaguchi [59] advocate for transforming disaster reports into disaster tales, which utilize multimedia elements, including images, to improve public understanding and preparedness for flooding events. The study by Kirakosyan [55] highlights the potential of picture story games to engage individuals, enhance understanding, and ultimately improve disaster preparedness outcomes. The three games in this study have also shown that connecting risk information and preparedness techniques with stories and day-to-day experiences enhances these outcomes.

### *5.2. Managerial Implications*

One of the main contributions is the involvement of teachers in designing games and leading them step-by-step through the game design process. Going through the game design process, they can use games and the game design process to enhance their pedagogy in the classroom in their future teaching endeavors. Additionally, as they know their students well, they were able to contribute substantially to developing games that appeal to the younger generations. The focus was on analog games to make these games available in areas without internet and computers.

The other main contribution is presenting three games for very specific disaster situations—landslides, fires, and earthquakes—and testing them with students. The main changes were summarized, and the final versions of the games were digitized and tested in another environment before they could be published. The ambition is to create professional games that can be translated into other languages and tested in other cultural environments. This is envisioned as the next step of this research.

The gamification of disaster preparedness and response is the future of disaster management. Academic efforts have been made to boost human resources for game design [60]. With growing access to the internet and virtual reality among younger generations, digital game applications are becoming part of daily life. This study has identified analog approaches to the gamification of disaster preparedness and response. Rural towns have better accessibility to the internet and mobile devices globally. The next step of this research is the application of emerging technologies in this field. This includes work using game engines for disaster prevention and mitigation that can be developed in the future using modern technologies, including artificial intelligence and virtual and augmented reality. They can be applied to evaluating rewards for emergency management education and disaster training [61] and developing generative artificial intelligence-based emergency training platforms.

The gamification of disaster preparedness can significantly improve community resilience, especially in rural areas. Schools offer an ideal environment for integrating these games, with high interest from students and teachers. Expanding this approach could contribute to building a more disaster-prepared society by targeting other vulnerable areas and populations globally.

Disaster preparedness, response, and recovery are continuum processes [62], reaching the last stretch of disaster preparedness and response by customizing and developing context-appropriate and socially relevant games. Distant geographies and vulnerable populations can be empowered by game-based learning. Populations in Nepal can significantly benefit from this knowledge about the most devastating disasters, including earthquakes, fires, and landslides. This study has shown that while digital games cannot be implemented in these remote locations, these analog games can be extended and contextualized to local needs.

### *5.3. Limitations of the Study*

One of the major limitations of this research is that only three games were developed to address three disasters. They all show promise, but further testing in other contexts and other disasters is needed. Further improvements will be made based on the most recent testing. Nepal's mountain, hill, and plain regions drastically differ in terms of disaster risk, culture, lifestyle, and geography. These games must be tested and vetted in these

very specific geographic regions. Similarly, they have to be implemented in coastal areas where the impacts of climate change are exacerbated. Games can be further developed by targeting multiple disasters. This may allow the flexibility of use in different geographic and disaster contexts. It may enhance the chances of adaption for these and other games by different governmental, non-governmental, and private agencies in the Global South. Besides preparedness and response, these games should also be tested for recovery planning efforts at the individual, household, and community levels, which may enhance the scope and adoption of these games by disaster management agencies.

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